

MILESTONE 2 – COVER PAGE

Team Number:

22

Please list full names and MacID's of all *present* Team Members

Full Name:	MacID:
Matthew Ho	Hom23
Ruidi Liu	liu127
Sharanya Srirangam	srirans
Mayar Aljayoush	Aljayoum

MILESTONE 2 (STAGE 1) – PRELIMINARY CONCEPTS

Team Number:

22

You should have already completed this task individually *prior* to Design Studio 15.

1. Copy-and-paste each team member's preliminary concept sketches on the following pages (1 sketch per page)
 - Be sure to indicate each team member's Name and MacID
2. Copy-and-paste each team member's description of their chosen input and output devices on the following pages (1 description per page)
 - Be sure to indicate each team member's Name and MacID

We are asking that you submit your work on both worksheets. It does seem redundant, but there are valid reasons for this:

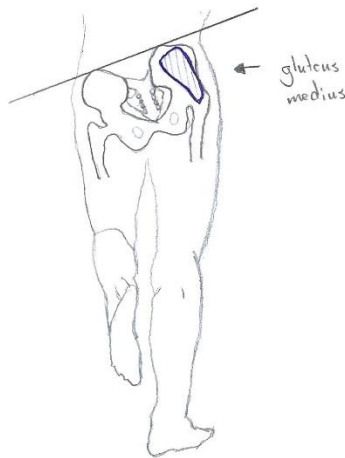
- Each team member needs to submit their preliminary concept sketches and description of their chosen input and output devices with the **Milestone Two Individual Worksheets** document so that it can be *graded*
- Compiling your individual work into this **Milestone Two Team Worksheets** document allows you to readily access your team member's work
 - This will be especially helpful when completing **Stage 2** of the milestone

Name: Matthew Ho

MacID: hom23

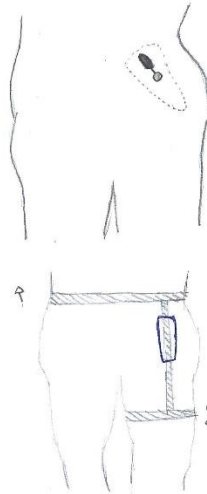
Preliminary Concept Solution #1

Hip Fracture (Trendelenburg Gait)



Internal view of the weak gluteus medius in Trendelenburg gait, which results in pelvic destabilization and abnormal hip movement. The gluteus medius is weak due to incident use post-surgery, and must be monitored to increase strength and return to normal gait patterns.

(Outlined gluteus medius)



Myoware muscle sensor will be placed on skin as indicated, with electrodes at the middle of the muscle body and parallel to the muscle length

← Sensor patch contains the EMG sensor

← Elastic-like bands (similar to athletic tape, or even suspenders) placed above hip and around thigh, and one strip connects the 2 sections while holding the sensor in place. The hip section prevents the mechanism from being pulled down (because it rests above and is tightened to the waist) and the thigh section prevents it from being pulled up. (indicated by ↑ and ↓ arrows)

- Full concept is drawn in the bottom right diagram. Other 2 diagrams depict Trendelenburg gait (left diagram) and how the sensor will be placed in relation to the muscle (top right diagram)
- **IMPORTANT (for marker):** This device is specifically geared towards Trendelenburg gait, which is a topic that we MAY focus on specifically, instead of muscle atrophy in general. This plan of action was discussed and approved by our TA in DP3 WK-2, and I was tasked with creating a concept sketch specifically for this topic. Thank you for understanding.

Team Number: 22

Name: <i>Matthew Ho</i>	MacID: hom23
Preliminary Concept Solution #1	

Type of information input from the physical environment

Muscle activity levels from the gluteus medius, the muscle associated with trendelenburg gait and hip fracture recovery. No other muscle activity should be measured on than the gluteus medius with this device.

Type of sensor required

MyoWare Muscle Sensor (EMG).

Type of output device(s) used for communication

Screen (any screen that can display numbers and text, ex. LED) **OR** coloured lights.

How output devices will be implemented

Ideally, a screen to display muscle activity levels over specific periods of time would be placed on the outside of the sensor path. This would communicate how much the user is activating that muscle, and whether they should increase, maintain, or decrease their gluteus medius activity in order to strengthen their weak side.

However, if a screen is not possible (depending on what 1P10 offers in this DP) then various coloured lights will be used on the same sensor path, and they will be assigned to different levels of activity. Depending on which light is lit, users can be informed of their muscle activity and adjust daily activity levels accordingly.

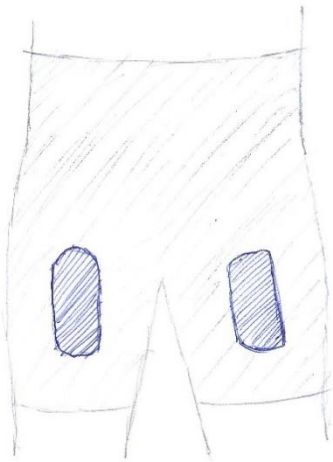
Name: Matthew Ho

MacID: hom23

Preliminary Concept Solution #2

Muscle Atrophy (for the general aging customer)

- ➔ When facing muscle deterioration, older patients should focus on maintaining the larger muscle groups, particularly those that play integral roles in whole-body mobility. Thus, this device should focus on upper to lower thigh emg sensors, to maintain walking ability.



- ← Flexible Under-Wear : "Shorts" that have materials similar to athletic under-garments, to allow for comfort, stretch and lightweight fabric. Will be worn over underwear and under, outer clothing. EMG sensors will be sewn/embedded into the shorts. They cannot be tampered with or removed.

- ← 4 emg sensors in total : 2 per leg, 1 front and 1 back. This way both legs can be measured and a more inclusive 3D understanding of thigh muscle activity can be achieved (however only 1 sensor can be used at a time,

☆ This device does not use individual straps, tapes, or contraptions to keep emg sensors in place. Instead, the use of an entire "shorts" piece of clothing allows for both secure hold of the sensors and full mobility of the body, without the hassle of straps or tapes.

Team Number: 22

Name: <i>Matthew Ho</i>	MacID: hom23
Preliminary Concept Solution #2	

Type of information input from the physical environment

Muscle activity of the large muscle group located in the upper to lower thigh. A single specific muscle does not have to be identified by the sensor each time, however the sensors should reside in a general area that can be repeatedly measured, in order to produce consistent and relative results.

Type of sensor required

MyoWare Muscle Sensor (EMG).

Type of output device(s) used for communication

Screen (any screen that can display numbers and text, ex. LED) **OR** coloured lights.

How output devices will be implemented

The implementation and communication of data collected by the sensor would be similar to concept sketch #1 (Trendelenburg gait). Ideally, a screen to display muscle activity levels over specific periods of time would be placed on the outside of the sensor path. This would communicate how much the user is activating that muscle, and whether they should increase, maintain, or decrease their muscle activity in order to maintain muscle strength.

However, if a screen is not possible (depending on what 1P10 offers in this DP) then various coloured lights will be used on the same sensor path, and they will be assigned to different levels of activity. Depending on which light is lit, users can be informed of their muscle activity and adjust daily activity levels accordingly.

Team Number: 22

Name: Sharanya Srirangam

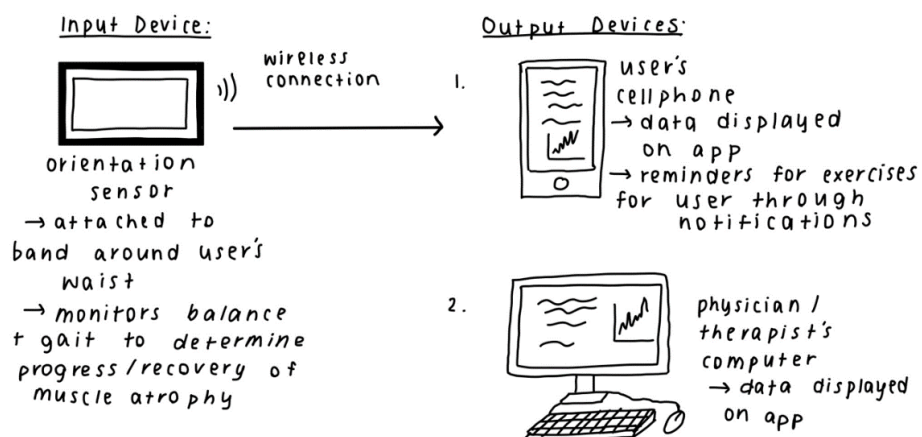
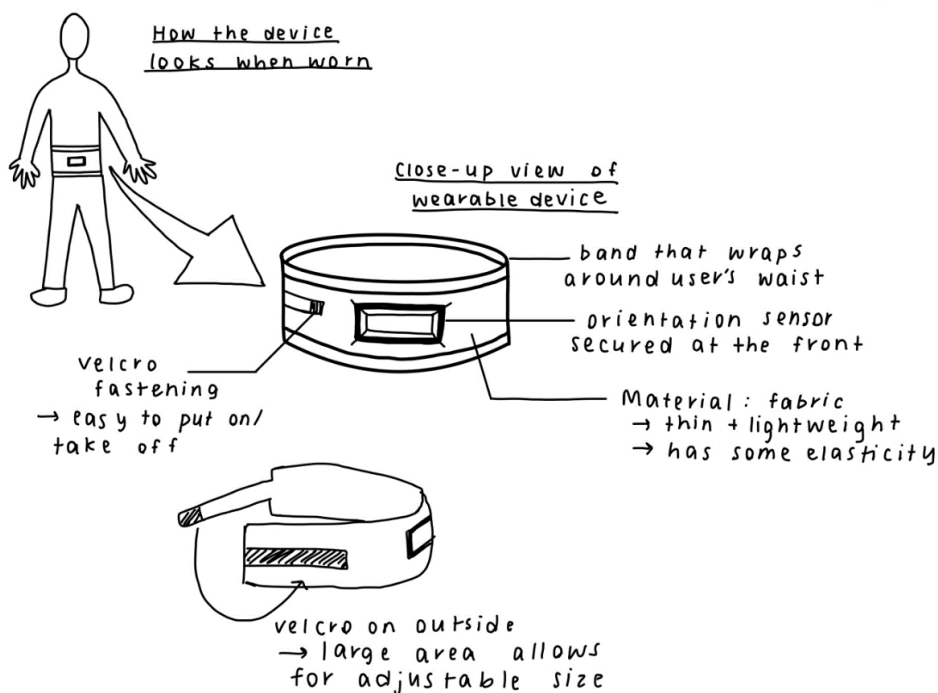
MacID: srirans

Preliminary Concept Solution #3

Insert screenshot(s) of your preliminary sketches below

Team #: 22 Name: Sharanya Srirangam mac ID : srirans

DP3-Preliminary concept sketch #1: balance and gait monitoring device for muscle atrophy



Team Number: 22

Name: <i>Sharanya Srirangam</i>	MacID: srirans
Preliminary Concept Solution #3	

Type of information input from the physical environment

Three-axis orientation in angular degrees

Type of sensor required

Orientation sensor

Type of output device(s) used for communication

Smartphone, Computer

How output devices will be implemented

- the sensor will be constantly monitoring the orientation of the user and sending this data wirelessly to their phone as well as the physician/therapist's computer
- the output will be in the form of a graph to provide a visual of the patient's progress in improving their balance
- when movement is not detected for longer periods of time, the phone app will remind the user to do their assigned exercises to improve balance, which will be done through notifications
- if the orientation's angular degrees are lower or higher than what the physician/therapist expects, they will be alerted through their computer app, which will allow them to change the exercise plan accordingly (For example, if the patient's balance is improving, the physician can try a more advanced exercise)
- if the maximum rotation angle that is expected is 20 degrees (threshold determined by doctor), and the patient exceeds 20 degrees for more than 3 minutes, the doctor will be notified
- if the patient's orientation data shows a lack of movement for a day or two (depending on the exercise plan), the patient will be sent a reminder through their app to improve their balance
- through the continuous monitoring and two output devices, the user's balance should improve, thus preventing falls that muscle atrophy patients often experience

Team Number: 22

Name: Sharanya Srirangam

MacID: srirans

Preliminary Concept Solution #4

Team #: 22 Name: Sharanya Srirangam mac ID: srirans

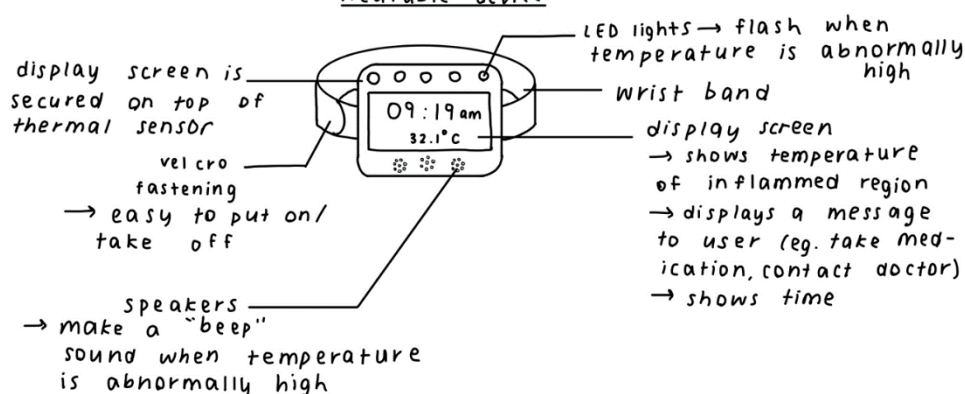
DP3-Preliminary concept sketch #2: inflammation monitoring device for muscle atrophy caused by rheumatoid arthritis (RA)

How the device looks when worn



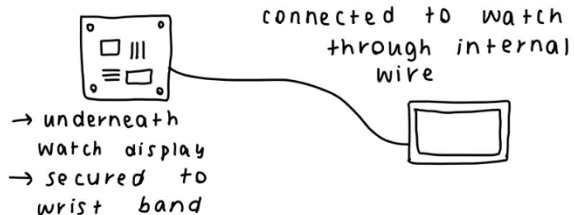
RA can affect several joints around the body. This concept sketch will focus on RA in the wrists and hands, as the device is designed as a watch.

close-up view of wearable device



Input Device:

thermal sensor



Output Devices:

1.  LED lights
2.  Speakers
3.  display screen

Team Number: 22

Name: <i>Sharanya Srirangam</i>	MacID: srirans
Preliminary Concept Solution #4	

Type of information input from the physical environment

Temperature (°C)

Type of sensor required

Thermal Sensor

Type of output device(s) used for communication

- LED lights
- speakers
- display screen of watch

How output devices will be implemented

- all devices will be connected through an internal wire (sensor will be secured underneath the display screen)
- an example of the implementation: when the temperature of the inflamed region (wrist or hand) exceeds 40°C for 10 minutes, the LED light will start flashing and there will be a “beep that is activated from the speakers (user can choose to disable one of these options)
 - the beeps/flashing lights will only last for 10 seconds
 - as well, a message will pop up on the display screen, reminding the user to take action to ease the inflammation (may be a reminder to take NSAIDS, ie: nonsteroidal anti-inflammatory drugs, which will ease the inflammation and pain or a suggestion to contact the user’s doctor)
- through the continuous monitoring of the area where the patient is experiencing rheumatoid arthritis, the thermal data can help ease their pain, thereby improving their lifestyle as a muscle atrophy patient

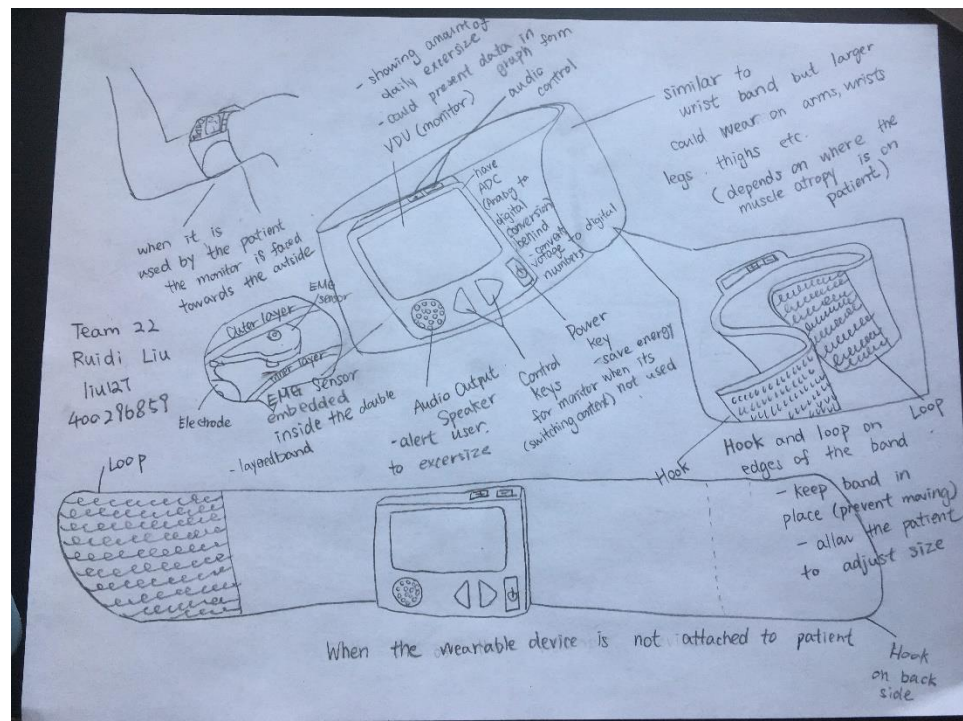
Team Number: 22

Name: Ruidi Liu

MacID:liu127

Preliminary Concept Solution #5

Insert screenshot(s) of your preliminary sketches below



Team Number:

22

Name: Ruidi Liu	MacID: liu127
Preliminary Concept Solution #5	

Type of information input from the physical environment

Filtered and rectified electrical activity of a muscle Electric potential of a muscle
--

Type of sensor required

EMG Sensors(MyoWare Muscle Sensor)

Type of output device(s) used for communication

Visual Display Units (VDU)/Monitor Audio Device (Speakers) Analog to Digital Conversion (ADC)

How output devices will be implemented

The monitor is directly on the wearable device connected with control keys that allow the user to switch content (daily exercise amount, exercise data etc.) on the screen. It is used for visually displaying data, informing how much exercise have the patient done in a day/week/month. The size of the screen should be around 2 to 3 times larger than a smart watch and it displays time and % battery left on the device.
--

The speakers are under the monitor on the wearable device. There are control keys connected to the speaker, allowing the user to adjust volume. It is mainly used for reminding users to exercise. It also informs the user how much exercise they have done at the end of the day.
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The monitor is directly on the wearable device connected with control keys that allow the user to switch content (daily exercise amount, exercise data etc.) on the screen.
--

The speakers are under the monitor on the wearable device. There are control keys connected to the speaker, allowing the user to adjust volume.
--

The ADC is hidden behind the screen, connecting both the monitor and the EMG sensors. The ADC is hidden behind the screen, connecting both the monitor and the EMG sensors. It converts measured voltages to digital data, which the data could be organized as a graph.

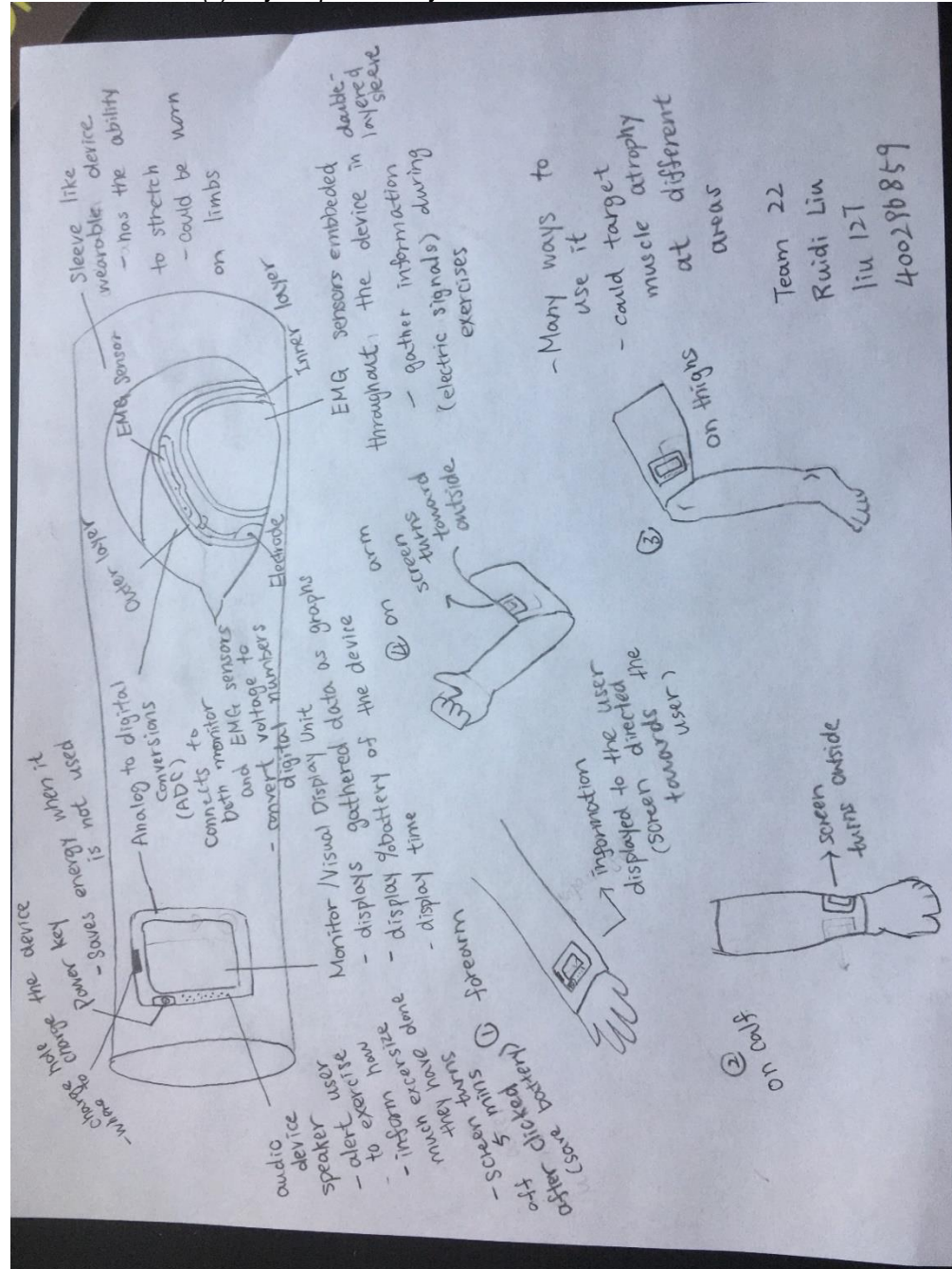
Team Number: 22

Name: Ruidi Liu

MacID:liu127

Preliminary Concept Solution #6

Insert screenshot(s) of your preliminary sketches below



Team Number: 22

Name: <i>Ruidi Liu</i>	MacID: liu127
Preliminary Concept Solution #6	

Type of information input from the physical environment

Filtered and rectified electrical activity of a muscle
Electric potential of a muscle

Type of sensor required

EMG Sensors (MyoWare Muscle Sensor)

Type of output device(s) used for communication

Visual Display Units (VDU)/Monitor
Audio Device (Speakers)
Analog to Digital Conversion (ADC)

How output devices will be implemented

The monitor is connected to a band (similar to a watch) and the band is sewed between the inside layer and outside layer of the double-layer sleeve (the wearable device). It is used for visually displaying data, time and % battery left on the device. The content on the monitor will change every 5 second, the screen turns off automatically after 5 mins.

*The space between the layer is not only for the monitor, but also where the EMG sensors are.

The speakers are beside the monitor, similar to how the speakers of our phone is beside the screen.

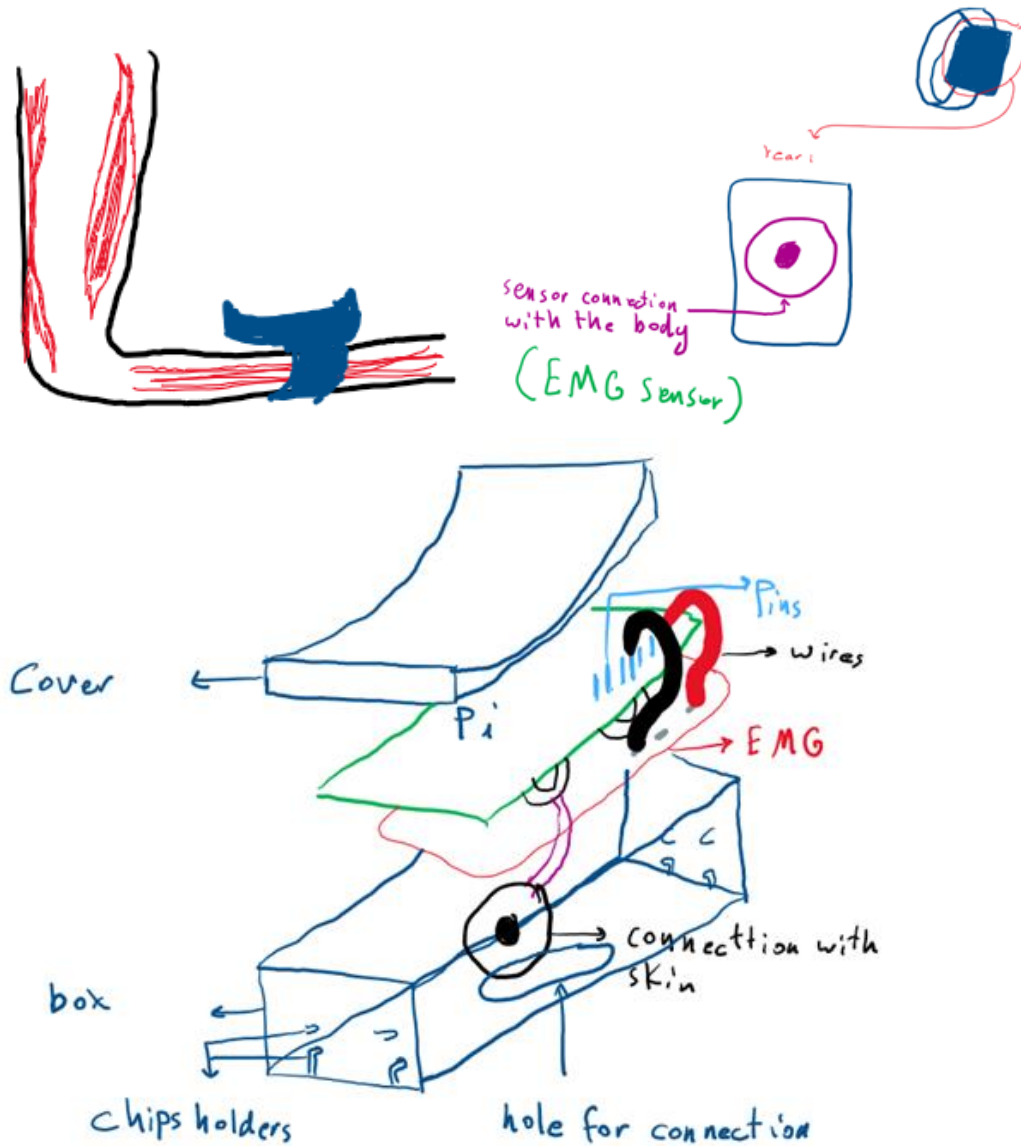
The ADC is hidden behind the screen, connecting both the monitor and the EMG sensors. It converts measured voltages to digital data, which the data could be organized as a graph.

Team Number: 22

Name: Mayar Aljayoush

MacID: Aljayoum

Preliminary Concept Solution #7



- The cover of the box has an arc shape to prevent collisions from damaging the chip and sensor so the impact can only be on the edges of the box
- The band is an elastic fabric that can stretch for the band to work on wrists, arms, thighs or even Waist

*If you are in a team of 5, please copy and paste the above on a new page

Team Number: 22

Name: <i>Mayar Aljayoush</i>	MacID: Aljayoum
Preliminary Concept Solution #7	

Type of information input from the physical environment

Sensing muscle movement to detect if the muscle is exercising enough.

Type of sensor required

EMG sensor

Type of output device(s) used for communication

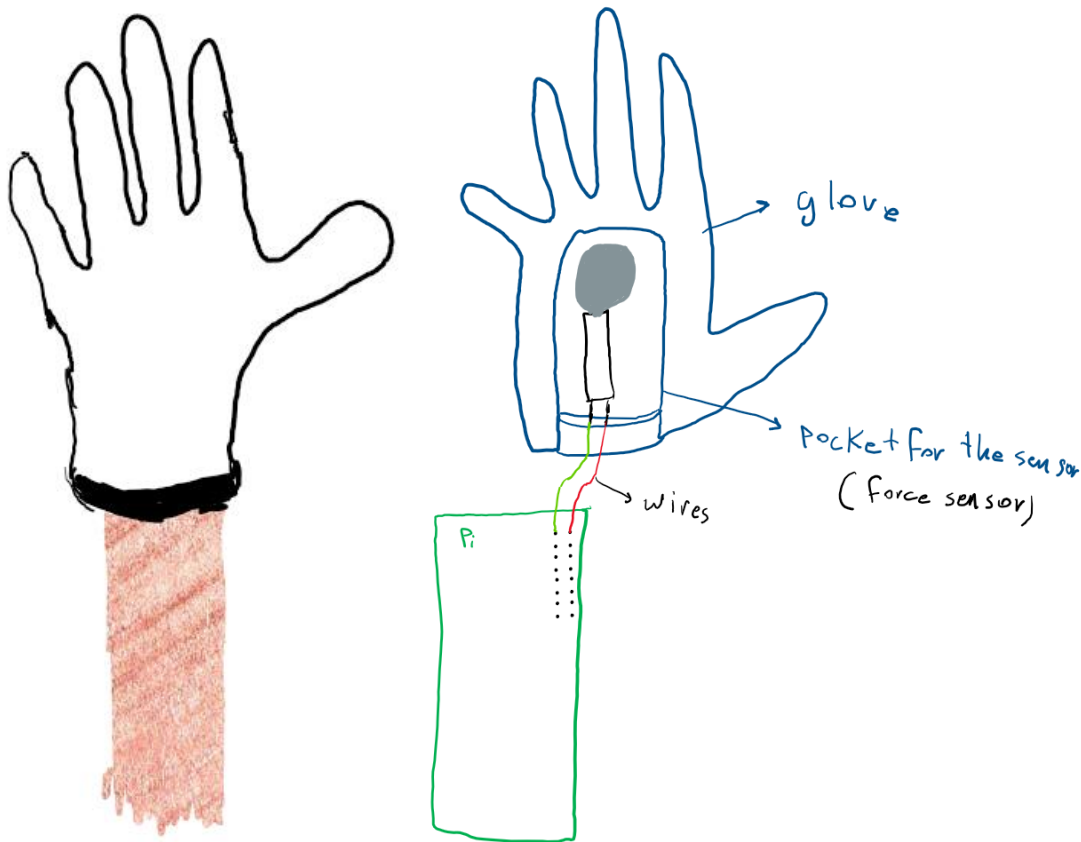
Send data to pi the pi would send a text message to the caretaker or patient that would include the length of the exercise, strength and a motivational letter.

How output devices will be implemented

Pi is in the device as well (or data receiving can be remote to decrease weight even more)

*If you are in a team of 5, please copy and paste the above on a new page

Team Number: 22

Name: <i>Mayar Aljayoush</i>	MacID: Aljayoum
Preliminary Concept Solution #8	
 The diagram shows two hand-drawn illustrations. On the left is a simple black outline of a right hand. On the right is a more detailed blue outline of a right hand wearing a glove. A grey oval on the palm represents a sensor, with a rectangular 'pocket for the sensor (force sensor)' below it. Two wires, one red and one green, extend from the pocket to a green rectangular box labeled 'Pi' (Raspberry Pi). The wires are labeled 'wires' with an arrow. Handwritten labels include 'glove' pointing to the blue outline and 'pocket for the sensor (force sensor)' pointing to the rectangular area on the palm.	

*If you are in a team of 5, please copy and paste the above on a new page

IBEHS 1P10 – Project Three: *Sense of Independence*

Team Number: 22

Name: <i>Mayar Aljayoush</i>	MacID: Aljayoum
Preliminary Concept Solution #8	

Type of information input from the physical environment

Grip strength

Type of sensor required

Force sensor

Type of output device(s) used for communication

Send data to pi the pi would send a text message to the caretaker or patient that would include the improvement rate/ weakness rate, advice to see a doctor if it's necessary and a motivational letter.

How output devices will be implemented

Data receiving can be remote using Wi-Fi or we it can e directly connected to pi

*If you are in a team of 5, please copy and paste the above on a new page

MILESTONE 2 (STAGE 2A) – PUGH MATRIX

Team Number: 22

1. As a team, evaluate your concept solutions in the table below

- List your Criteria in the first column
 - You should include a minimum of 5 criteria
- Fill out the table below, comparing your designs against a baseline concept (the decision of a baseline concept is arbitrary and entirely up to your team to decide)
 - Replace “Concept 1”, “Concept 2”, etc. with more descriptive labels (e.g., a distinguishing feature or the name of student author)
 - Indicate a “+” if a concept is better than the baseline, a “–” if a concept is worse, or a “S” if a concept is the same

	<i>Matthew 1</i>	<i>Matthew 2</i>	<i>Sharanya 1</i>	<i>Sharanya 2</i>	<i>Ruidi 1</i>	<i>Ruidi 2</i>	<i>Mayar 1</i>	<i>Mayar 2</i>
<i>Reduce muscle atrophy</i>	S	S	–	–	D	S	S	S
To improve quality of life	S	S	S	S	A	S	S	S
Be secure	–	+	S	S	T	S	S	S
Be reliable	S	S	S	S	U	S	S	—
Non-irritating	S	S	S	S	M	S	S	S
Hypoallergenic	S	S	S	S	D	S	S	S
Easy to put on/take off	–	+	S	+	A	–	+	S
Communicate data to the user/caretaker	–	–	+	S	T	S	+	+
Small size	+	–	S	+	U	S	+	S
Lightweight	+	S	S	+	M	S	+	S
Total +	2	2	1	3	-----	1	4	1
Total –	3	1	1	1	-----	0	0	1
Total Score	-1	+1	0	+2	-----	-1	+4	0

IBEHS 1P10 – Project Three: *Sense of Independence*

*For a team of 5, click the top-right corner of the table to “Add a New Column”

2. List the concepts you will consider for your detailed concept evaluation

- | |
|--|
| <ul style="list-style-type: none">• Matthew 2 (shorts-like wearable design targeting general muscle atrophy - EMG sensor)• Sharanya 1 (Orientation belt- orientation sensor)• Sharanya 2 (Inflammation watch – temperature sensor)• Mayar 1 (EMG band targeting general muscle atrophy- EMG sensor) |
|--|

MILESTONE 2 (STAGE 2B) – DECISION MATRIX

Team Number: 22

1. As a team, establish a **weighting factor** for each criterion to establish a *relative importance of criteria*. Complete this weighting factor calculation in the table below

→ List your Criteria in the first column *and* the first row

- You should include a minimum of 5 criteria

→ Moving row-by-row, indicate whether a particular criterion is preferred over another. For example:

- If *Criteria 1* is preferred over *Criteria 2*, assign a 1. Otherwise, assign 0
- Similarly, if *Criteria 1* is preferred over *Criteria 3*, assign a 1. Otherwise, assign 0

	Reduce muscle atrophy	To improve quality of life	Be secure	Be reliable	Non-irritating	Hypoallergenic	Easy to put on/take off	Communicate data to the user/caretaker	Small size	Lightweight	Score
Reduce muscle atrophy		0	1	1	1	1	1	1	1	1	8
To improve quality of life	0		1	1	1	1	1	1	1	1	8
Be secure	0	0		0	1	1	0	0	1	1	4
Be reliable	0	0	0		1	1	0	0	1	1	4
Non-irritating	0	0	0	0		0	0	0	0	0	0
Hypoallergenic	0	0	0	0	0		0	0	0	0	0
Easy to put on/take off	0	0	0	0	1	1		0	1	1	4
Communicate data to the user/caretaker	0	0	1	1	1	1	1		1	1	7
Small size	0	0	0	0	1	1	0	0		0	2
Lightweight	0	0	0	0	1	1	0	0	0		2

*If there are more than 7 criteria for consideration, click the top-right corner of the table to “Add a New Column” and the bottom-left corner of the table to “Add a New Row”

IBEHS 1P10 – Project Three: *Sense of Independence*

2. As a team, any concepts you have selected for further consideration (based on the outcome of the Pugh Matrix) should be evaluated in the table below

- List your Criteria in the first column
- List your **weighting factor** in the second column
- Fill out the table below, comparing your concept solutions against the design criteria
 - Replace “Concept 1”, “Concept 2”, etc. with more descriptive labels (e.g., a distinguishing feature or the name of student author)
 - Measure each concept solution against criteria on a scale from 1 (poor) to-5 (excellent), which serves as a **rating factor (R.F.)**
 - For each concept solution, calculate the **weighted rating factor (W.R.F.)** for each design criterion, which is the product of the weighting factor and rating factor
 - Add the **weighted rating factors** for each concept solution and compare results

	Weighting Factor	EMG Shorts		Orientation Belt		Inflammation Watch		EMG Band	
		R.F.	W.R.F.	R.F.	W.R.F.	R.F.	W.R.F.	R.F.	W.R.F.
Reduce muscle atrophy	0.2	4	0.8	3	0.6	2	0.4	5	1
To improve quality of life	0.2	4	0.8	3	0.6	4	0.8	5	1
Be secure	0.1	5	0.5	5	0.5	5	0.5	5	0.5
Be reliable	0.1	5	0.5	5	0.5	5	0.5	5	0.5
Easy to put on/take off	0.1	5	0.5	4	0.4	5	0.5	5	0.5
Communicate data to the user/caretaker	0.2	1	0.2	5	1	5	1	5	1
Small size	0.05	1	0.05	3	0.15	5	0.25	4	0.2
Lightweight	0.05	2	0.1	3	0.15	5	0.25	4	0.2
Total	1.0	27	3.45	31	3.9	36	4.2	38	4.9

*If there are more than 4 concept solutions for consideration, click the top-right corner of the table to “Add a New Column”

3. Indicate the concept you have selected to pursue for further development and testing (include justification)

The concept that is selected to pursue for further development and testing is the EMG band. The EMG band has versatility as it could target different muscle groups on the user’s body. It is secure, small size and lightweight without limiting the user’s movement. Since our customer demographic is elderly people, it is necessary for the EMG band to be easily equipped. Furthermore, the EMG band specifically monitors muscle activity which reduces the muscle atrophy effectively, compared to other concept solutions.